

2023



Autodesk Boosts Startup Performance by 50% Using Amazon ECS

Learn how Autodesk improved performance of long-running hydraulic modeling simulations using Amazon ECS and AWS Fargate.

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50% reduction

in simulation startup time

Minimized

operational overhead

Improved

availability and security

Overview

When programs are loading, minutes can fly by. For Autodesk, these wait times were a significant barrier to understanding how water behaves in certain systems, the wait often isn't worth the result, leading to abandoned projects.

To help its customers with long-running simulations, Autodesk moved to cloud-based solutions using Amazon Web Services (AWS). These would bring faster, more efficient, and more cost-effective software-as-a-service versions of two of its water simulation products: InfoWorks ICM and InfoWorks ICM-B. The result so resulted in an improved customer experience and the ability to start up and roll out new features easily.



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Opportunity | Using Serverless Compute to Build Cloud Solutions for the Water Industry

Founded in 1982, Autodesk is a software company that provides products and services across industries, including product design, manufacturing, media and entertainment, and architecture. In the engineering industry, two of the software products that Autodesk offers are InfoWorks ICM and InfoWorks WS Pro. These help municipalities and consultants to model complex simulations to better understand the behavior of water in collection, infrastructure, and clean water systems.

Previously, customers had two deployment options for these products. They could elect to run everything, both short and long simulations, on their own desktop computers, but this led to many customers scaling back or abandoning their projects because of time and compute-power limitations. Another option was for customers to scale up compute power by building on-premises infrastructure, but building out large-scale on-premises infrastructure was time-consuming and required significant effort and resources. Customers would need to plan projects, purchase servers, set up servers, and then install the software to be run. A project like this could easily take a full year and cost more than \$100,000. And then the expensive servers would be left idle when there were no active simulations to run.

To make it faster and simpler for customers to run simulations, Autodesk set out to create cloud-based versions of its products. “We wanted to provide the flexibility to run as many simulations as they wanted, when they wanted,” says Boaz Brudner, head of Innovyze software-as-a-service engineering at Autodesk. “The goal was to democratize accessibility to large-scale compute power to anyone who uses our products.” Autodesk’s solution went further than a simple lift-and-shift migration, which would have left it with servers to manage and maintain. Instead, the team rebuilt 90 percent of the simulation services for InfoWorks ICM and InfoWorks WS Pro from the ground up in the cloud using AWS serverless services.



Solution | Reducing Water Simulation Startup Time by 50% Using Amazon ECS and AWS Fargate

For Autodesk, using a serverless-first approach was of utmost importance so that its team could off-load the management of servers, reduce operational overhead, and focus valuable time on innovation. Autodesk rebuilt its products using [Amazon Elastic Container Service](#) (Amazon ECS)—a fully managed container orchestration service that simplifies deployment, management, and scaling of containerized applications, with [AWS Fargate](#), a serverless compute engine that developers can use to run containers without having to manage servers or clusters. “Building completely serverless solutions minimizes both our overhead cost for operations and our exposure in terms of security and removes the need for us to deal with patching and updating servers,” says Brudner. Although developers were able to run some CPU-intensive workloads on AWS Fargate, they ran their GPU instances using Amazon ECS on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload.

As part of its serverless architecture, the team also implemented [AWS Lambda](#), a serverless, event-driven compute service that developers use to run code for virtually any type of application, and [Amazon EventBridge](#), a serverless event bus used to receive, transform, route, and deliver events.

Autodesk wanted to make sure that customers noticed a performance boost when switching from running simulations on premises to the cloud. For this, Autodesk used [Seekable OCI](#) (SOCi), a technology that is open sourced by AWS and supported by AWS Fargate and that speeds processing time by starting up container tasks without waiting for the entire container image to load. Using SOCi led to a 50 percent decrease in startup time for some simulations running on Amazon ECS instances. “This helps our application to scale out faster, making it possible for us to quickly serve increased user demand and save on costs by reducing idle compute capacity,” says Ocie Mitchell, senior principal engineer at Autodesk.

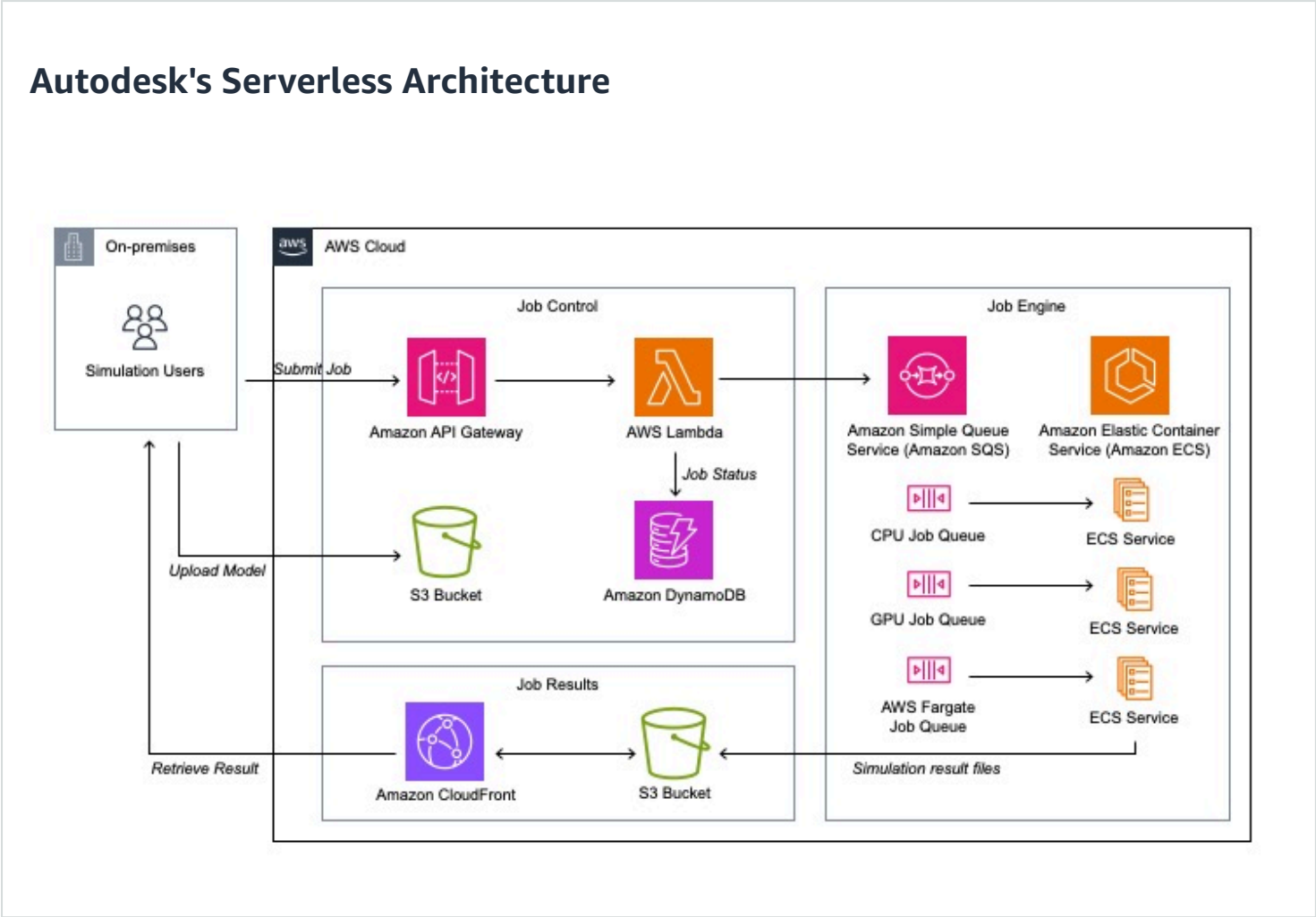
Improved customer experience didn’t end with faster startup times. “Using cloud-connected products, customers can focus on the problems they’re solving, not the infrastructure they’re using,” says Mitchell. Onboarding customers is quicker and simpler than before, and customers can begin running as many simulations as they want as soon as they buy the product. Using Amazon ECS, Autodesk’s servers now automatically scale to demand, so it maintains high availability for its customers. They can run a high number of long simulations in parallel when before they would have refrained due to limited resources. The high-performance compute power of the software also saves customers time—simulation startup overhead was reduced by 50 percent.



software-as-a-service model reduces the need for idle instances. Customers can reduce the overhead costs of running simulations.

Using a serverless-first approach benefited not only Autodesk's customers but their own development teams as well. Now, they can make changes to the software faster than before—without interrupting customer experience. “The ability to innovate in the cloud is a huge differentiator for us,” says Brudner. “Using AWS, we can update, improve, and introduce new capabilities on a daily basis without making major software releases or requiring customers to update their software or servers. We update our services, and our customers have new capabilities right away.”

Architecture Diagram



Outcome | Building Cloud-Connected Capabilities on AWS

Autodesk successfully transformed its desktop offerings into cloud-connected solutions. On the cloud, Autodesk now offers its customers better, faster performance at a fraction of the previous cost. In addition, it can roll out new features quickly and easily.

Autodesk has begun to transition more of its desktop products, including InfoWater Pro and InfoDrainage, to the cloud-connected model so that customers can benefit from more capabilities and use cases. “The most important thing is to focus on what customers are trying to do and how we can elevate their capabilities,” says Brudner. “We are going to double down on customer needs because that is what drives u

About Autodesk

Autodesk Inc. provides software products and services across several industries to empower innovators with design. Its InfoWorks ICM and InfoWorks WS Pro software help architects, engineers, and construction professionals model complex water simulations.

AWS Services Used

Amazon ECS

Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service that simplifies your deployment, management, and scaling of containerized applications.

[Learn more »](#)

AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers. AWS Fargate is compatible with both Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS).

[Learn more »](#)

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)

Amazon EventBridge

Amazon EventBridge is a serverless service that enables developers to create scalable event-driven applications by routing events between your own applications, third-party software-as-a-service (SaaS) applications, and AWS services.



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